

NMEC591 - Research Methodology and Statistics

Lecture 0: Why Learn?

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The Beginning: Curiosity

Every research journey begins with a simple moment:

- Something does not behave as expected.
- A result looks surprising or inconsistent.
- A question refuses to go away.

Research methodology teaches us how to turn curiosity into structured inquiry.

The First Mistake

Many beginners rush to find answers.

- They collect data without a clear question.
- They try methods without understanding assumptions.
- They reach conclusions before evidence is ready.

Methodology exists to slow us down—and guide us correctly.

Learning to Ask the Right Question

A well-posed research question:

- Defines what is known and what is unknown.
- Is precise, testable, and meaningful.
- Connects theory, observation, and application.

Research methodology teaches how to shape vague problems into clear questions.

Designing the Path Forward

Once the question is clear, the path matters.

- What data should be collected?
- What variables should be controlled?
- What tools are appropriate?

Methodology helps design a study that can actually answer the question.

Meeting the Data

Data rarely speaks clearly at first.

- Measurements contain noise and uncertainty.
- Samples may be biased or incomplete.
- Patterns can be misleading.

Statistical thinking provides the language to interpret data responsibly.

Avoiding False Conclusions

Without methodology:

- Correlation is mistaken for causation.
- Random variation is treated as discovery.
- Results cannot be reproduced.

Research methodology protects us from drawing confident but incorrect conclusions.

Ethics and Responsibility

Research affects people, technology, and society.

- Data must be reported honestly.
- Methods must be transparent.
- Results must be reproducible.

Methodology embeds ethics into the research process.

Communicating the Story

A discovery has value only if it is understood.

- Clear structure improves credibility.
- Logical flow strengthens arguments.
- Proper reporting enables peer evaluation.

Research methodology teaches how to present, not just compute.

The Bigger Picture

Research methodology is not just for researchers.

- Engineers use it to validate designs.
- Scientists use it to test theories.
- Professionals use it to make data-driven decisions.

It is a way of thinking, not just a set of tools.

The End—and the Beginning

By studying research methodology, you learn to:

- Ask better questions.
- Trust your conclusions.
- Contribute knowledge, not just results.